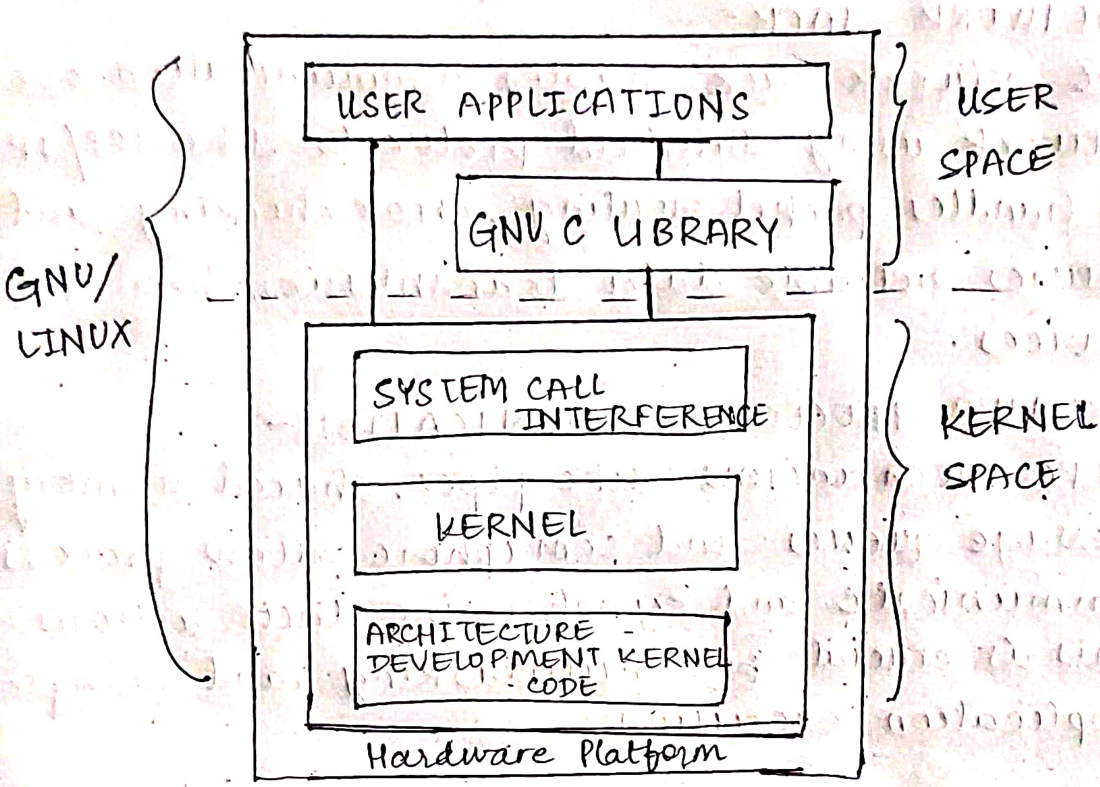
NAME - ARUSHI PRIYA
REGD NO - 2141016384

TASK 1

Kernel Architecture diagram



SHORT DESCRIPTIONS

1) PROCESS SCHEDULER -

The scheduler manages process execution by determining which process runs at any given time. It ensures fairness and efficiency, supporting various scheduling algorithms for multitasking and time sharing

2) FILE SYSTEM -

The file system manages how data is stored, organized and retrieved on storage devices. It supports various formats like ext4, FAT, and ~~NA~~ NTFS & provides abstraction for file operations via system call

3) MEMORY MANAGEMENT UNIT -
The component handles the execution allocation and deallocation of memory to processes. It support virtual memory, page swapping and memory protection to ensure stability and performance

4) NETWORK STACK -
The network stack enables communication over network using standard protocols like TCP/IP. It handles packet routing, error checking and ensures reliable data transmission between devices.

5) INTER-PROCESS COMMUNICATION -
IPC mechanisms like pipes, shared memory, message queues and semaphore allow process to communicate and synchronize their actions. This is crucial for multitasking and complex application execution.

6) SYSTEM CALL INTERFERENCE -
The SCI provides a bridge for user space applications to interact with kernel functions. It translates user requests into kernel-level instructions for resources access and control

7) KERNEL MODE AND USER MODE -
The Linux kernel operates in kernel mode with unrestricted access to all system resources, whereas user applications run in user mode with limited access. This separation enhances system security and stability.